

Supplementary Material

Does occluding the ear free loudness budget for high-frequency gain?

A modeling study of standard audiograms

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S1. Validation of the C++ Loudness Engine

The AUDMOD specific-loudness engine, the `bramslow2004` implementation in AMT 1.6.0 (Majdak et al., 2022), was ported to C++ using `Rcpp` and validated at three levels: port fidelity, input path consistency, and absolute reference-free behavior.

First, port fidelity was established by comparing the C++ engine to the AMT `bramslow2004` (AUDMOD) implementation stage by stage. Across 23 test cases, the maximum relative discrepancy at any stage of computation was 5.71×10^{-15} . This comparison establishes that the C++ port reproduces AUDMOD; it does not bear on whether AUDMOD itself is accurate.

Second, the input path for spectrum construction feeding the engine was verified. The two independent routines used to compute normal-hearing speech loudness in this work return identical values at 50, 65, and 80 dB SPL (3.1577, 9.0333, and 22.3887 sones, respectively). Evaluating the 65 dB SPL speech spectrum on a 10–23990 Hz grid without renormalization yields 9.0333 sones, whereas evaluation on a 20–15000 Hz grid with renormalization yields 9.0317 sones; this difference of 0.0016 sones shows that grid range and renormalization are immaterial. Supplying a precomputed reference versus letting the wrapper build one dynamically yields identical results for three sloping test audiograms (13.2297, 20.7120, and 16.6072 sones).

Third, the absolute behavior of the model was verified using reference-free checks, since the port and AMT would share any error. Loudness correctly evaluates to zero below threshold: for normal hearing, a 1 kHz tone remains silent up to +5 dB SPL, and for a test audiogram with an 80 dB HL threshold at 4 kHz, a 4 kHz tone remains silent up to 60 dB SPL. Recruitment is present and ordered by the 4 kHz threshold across five test audiograms, including a reverse-slope audiogram with a 20 dB HL threshold at 4 kHz. (These checks used separate test audiograms; they concern the engine, not the audiograms analyzed in the main text.) In normal hearing, the loudness growth for a 1 kHz tone is 1.75 to 1.91 per 10 dB above 40 phons, which characterizes the model as mildly compressive relative to the textbook expectation of a doubling per 10 dB.

The absolute sone scale is inherited from AUDMOD and has not been independently verified against loudness judgments. An attempt to check the 1 kHz / 40 dB SPL = 1 sone anchor was inconclusive: the R wrapper interpolates the input spectrum linearly in dB across a log-frequency grid, which is appropriate for the smooth, densely sampled speech spectra used throughout this work but cannot faithfully represent a pure tone. The smooth-spectrum control was stable at 9.0333 sones across every input grid spacing tested, so the analyses reported in the main text are unaffected. We do not claim the absolute scale is either correct or incorrect.

S2. The Open-NL Objective and Optimizer Settings

Objective

Open-NL optimizes six gain shifts, $s = (s_1, \dots, s_6)$, one at each octave frequency from 250 to 8000 Hz. Each shift is clamped to the range -60 to $+30$ dB and added to the heuristic

37 prescription g_k^h (Section S3), and the resulting gain is limited to the range from the minimum-
 38 gain floor g_{floor} (0 or -10 dB) to 80 dB:

$$g_k = \min(80, \max(g_{\text{floor}}, g_k^h + s_k)).$$

39 The optimizer minimizes

$$J(s) = -100 \cdot \text{SII}(g) + P_{\text{anchor}} + P_{\text{rough}} + P_{\text{loud}} + P_{\text{bound}} + P_{\text{SPL}} + P_{\text{ABG}},$$

40 where the SII is the complete desensitized SII described below, so that one point of J
 41 corresponds to 0.01 SII. The three penalties that shape the reported solutions are

$$P_{\text{anchor}} = 0.1 \sum_k |s_k|, \quad P_{\text{rough}} = 0.001 \sum_{k=1}^5 (g_{k+1} - g_k)^2, \quad P_{\text{loud}} = 2000 \max(0, N - N_{\text{cap}}),$$

42 where N is the AUDMOD loudness of the aided speech spectrum in sones and N_{cap} is the
 43 loudness cap. In the anchor-off analyses the weight 0.1 in P_{anchor} was set to zero; nothing else
 44 was changed.

45 The remaining terms are safeguards that were inactive or rarely engaged in these
 46 analyses. $P_{\text{bound}} = 1000 \sum_k [\max(0, s_k - 30)^2 + \max(0, -s_k - 60)^2]$ acts on the unclamped
 47 search variables: because the shifts applied to the gain are clamped, the objective is flat beyond
 48 that range, and the penalty keeps the search from drifting there. $P_{\text{SPL}} = 2000 \max(0, L - 110)$
 49 penalizes an overall aided output L above 110 dB SPL. $P_{\text{ABG}} = \sum_k \max(0, s_k)^2$ is applied only
 50 to audiograms with an air-bone gap (none in this study) and penalizes gain above the heuristic
 51 prescription, which already incorporates the conductive correction. Two further terms, not
 52 shown, constrain the gains at 50 and 80 dB SPL relative to the 65 dB SPL solution (ordering of gain

across levels and a maximum compression ratio); because all optimizations here were performed at 65 dB SPL, these were not active.

SII Evaluation within the Objective

The octave-frequency gains are interpolated linearly in log frequency to the 21 bands of the ANSI S3.5-1997 critical-band procedure and applied to the standard speech spectrum for normal vocal effort, scaled to 65 dB SPL. The SII is then computed with the standard band-importance function, the standard level distortion factor, and an external noise floor of -50 dB, which leaves hearing threshold and self-speech masking as the effective disturbances. Desensitization is applied to each band's audibility K_i through two functions of the band's sensorineural threshold T_i (dB HL):

$$m_i = \frac{1}{1 + e^{0.075(T_i - 66)}}, \quad p_i = \frac{T_i}{8} - 15.$$

The complete correction, which is the optimizer's objective, is the power mean $K'_i = (K_i^{p_i} + m_i^{p_i})^{1/p_i}$; the ANSI SII, reported alongside it, uses K_i unchanged. For the thresholds used here (0 to 110 dB HL), p_i lies between -15 and -1.25 . At mild to moderate thresholds, where p_i is strongly negative, the complete correction behaves approximately as $\min(K_i, m_i)$: once a band's audibility exceeds its desensitization ceiling m_i , further audibility earns little credit. At severe to profound thresholds the transition is more gradual. Negative insertion gain lowers the speech level in the SII calculation, as it does in the loudness model.

Optimizing the ANSI SII

To represent a fitting tuned to the uncorrected index that real-ear systems display, the tightest-budget sweep and the iso-loudness control were repeated with the optimizer maximizing

the ANSI SII, everything else unchanged. Tables S1 and S2 give the floor effect of these solutions under the ANSI SII (what the display would show) and under the complete SII.

Table S1. Floor Effect of ANSI-Optimized Solutions: Audiogram Family Sweep at the 0.5-Sone Budget

Audible edge	ANSI (on)	Complete (on)	ANSI (off)	Complete (off)
1000 Hz	+0.079	+0.013	+0.087	+0.014
1500 Hz	+0.027	+0.004	+0.038	+0.005
2000 Hz	+0.010	+0.004	+0.017	+0.001
3000 Hz	−0.001	−0.001	+0.002	−0.001

Note: Means over the four slopes at each edge. With the anchor penalty all 16 pairs matched in loudness; without it, 5 did.

Table S2. Floor Effect of ANSI-Optimized Solutions: Iso-Loudness Control

Profile	ANSI (on)	Complete (on)	ANSI (off)	Complete (off)
N1	+0.007	+0.005	+0.013	+0.007
N2	+0.008	+0.007	0.000	−0.005
N3	+0.003	+0.002	+0.007	+0.008
N4	−0.005	−0.001	−0.001	−0.003
N5	0.000	0.000	0.000	0.000
N6	0.000	0.000	0.000	0.000
N7	0.000	0.000	0.000	0.000
S1	+0.006	−0.001	+0.023	0.000
S2	+0.011	−0.001	+0.027	−0.001
S3	−0.001	0.000	0.000	0.000

Note: Only N1, N2, S1, and S2 received negative gain; elsewhere the floor did not engage and nonzero values reflect search variability. Without the anchor penalty the −10 dB solution for S1 left 1.27 sones of the budget unspent; all other pairs matched to within 0.04 sones.

In the sweep, the ANSI-optimized solutions matched in loudness had no lower penalized objective at -10 dB than at 0 dB, so the search converged as it did for the complete objective.

Loudness Evaluation within the Objective

The aided spectrum at the six octave frequencies (the speech spectrum level plus gain) is interpolated to a 10 Hz grid from 20 to 15000 Hz, with skirts of 24 dB/octave below 250 Hz and above 8000 Hz, and rescaled to the overall aided level. Any air-bone gap is subtracted before the spectrum enters the AUDMOD engine. The audiogram supplied to AUDMOD is the sensorineural threshold (threshold minus air-bone gap), interpolated from the octave frequencies to AUDMOD's 13 audiogram frequencies (125 to 12500 Hz); the uncomfortable loudness level is left at the AUDMOD default of 120 dB HL, and the engine runs at a sampling rate of 32 kHz with an 8192 -point transform. The cap N_{cap} was set by each analysis: the listener's unaided loudness plus the budget in the audiogram family sweep, and the loudness of the NAL-NL2 prescription in the iso-loudness control.

Search Settings

The objective is minimized with the Nelder-Mead simplex method (R `optim`, maximum 800 iterations, other settings at their defaults) from multiple starting points. The unshifted heuristic ($s = 0$) is evaluated first and retained if no search improves on it. The first search starts from a deterministic point: in each octave band, a nominal shift toward the gain that would place a 65 dB SPL input 10 dB above threshold, limited to the range 0 to 20 dB. Each further start adds independent uniform jitter of ± 5 dB to every parameter. The solution with the lowest J across all starts is returned, so the reported solution is penalized-optimal and need not have the highest SII.

The random-number seed is set from the audiogram and the input level (100 times the sum of the thresholds, plus the level), so repeated runs are bit-identical and cannot be used to estimate variability, and a run with more starts repeats the starting points of a run with fewer. The number of starts can alter the prescribed gains, because the objective is non-convex; the analyses used 20 starts, with 40 for the one tightest-budget pair noted in the main text.

S3. The Open-NL Heuristic Prescription

The heuristic prescription g_k^h is a rule-based insertion-gain rule built into Open-NL; it serves as the anchor for P_{anchor} and as the origin of the search. At a 65 dB SPL input, and with the settings used here (fully occluded coupling, male, experienced bilateral user, no measured loudness discomfort levels, no dead-region or distortion information, severe-loss booster disabled), the gain at frequency f is computed as follows.

1. **Base gain.** $0.46T_{\text{SN}} + C(f) \cdot T_{\text{SN}}/T$, where T is the threshold, T_{SN} its sensorineural part, and $C(f)$ equals -8 , -1 , $+3$, and $+1$ dB at 250, 500, 1000, and 2000 Hz and 0 dB from 3000 Hz up, interpolated in log frequency.
2. **Reverse-slope correction.** When the mean sensorineural threshold at 500 Hz and below exceeds the mean at 2000 Hz and above by more than 15 dB, $C(f)$ below 1000 Hz is blended toward -10 dB, fully so when the difference reaches 35 dB, with a weight that falls linearly in log frequency from 1 at 250 Hz to 0 at 1000 Hz.
3. **Slope-dependent low-frequency penalty.** When the mean sensorineural threshold at 2000 Hz and above exceeds the mean at 500 Hz and below by more than 15 dB, low-frequency gain is reduced by up to 15 dB, with the same 250–1000 Hz taper. The reduction grows over the

next 20 dB of slope and is phased out as the high-frequency mean rises from 70 to 95 dB HL, so that listeners with profound high-frequency loss retain low-frequency gain.

4. **High-frequency soft limit.** Gain in excess of $30 + 0.4 \max(0, T_{\text{SN}} - 60)$ dB is halved at frequencies where the sensorineural threshold is more than 25 dB worse than the best threshold at or below 1000 Hz. The limit is phased in between 2000 and 4000 Hz and, in threshold, over a further 20 dB.

5. **Bandwidth weighting.** The gain is multiplied by 0.7 at 250 Hz, 1.0 from 500 to 4000 Hz, 0.8 at 6000 Hz, and 0.5 at 8000 Hz.

6. **Conductive correction.** 75% of the air-bone gap is added.

7. **Limits.** The gain is floored at -10 dB. A maximum-output limit is then applied (it does not bind for any profile at a 65 dB SPL input), and the resulting gain is floored at 0 dB. This final 0 dB floor is why, as noted in the main text, any gain below 0 dB is a penalized shift.

Open-NL's compression stage leaves the 65 dB SPL gain unchanged by construction, and demographic and coupling corrections are zero at the settings used, so the rule above is the complete heuristic for these analyses.

S4. Robustness of the Floor Effect

Two further analyses, both optimizing the complete SII at 65 dB SPL with 20 random starts, tested whether the floor-effect results depend on the loudness at which the comparison is made and on the random starts of the search.

Floor Effect at the Normative Ceiling

For the four standard audiograms with audible low-frequency speech, the optimization was repeated with the loudness cap set to L_{cap} , the loudness of unaided speech for a 0 dB HL listener (9.03 sones). Table S3 gives the results. No solution gave any band negative gain, with or without the anchor penalty, so the floor did not engage and the differences between floors reflect search variability alone.

Table S3. Floor Effect with the Loudness Cap at the Normative Ceiling (9.03 Sones)

Profile	Anchor	Budget	NAL-NL2 use	Sones (0 dB)	Sones (−10 dB)	SII (0 dB)	SII (−10 dB)	Floor effect
N1	on	2.53	0.54	9.03	9.03	0.890	0.896	+0.006
N1	off	2.53	0.54	8.93	9.01	0.915	0.915	0.000
N2	on	5.94	1.75	6.43	6.26	0.836	0.824	−0.012
N2	off	5.94	1.75	6.49	6.20	0.846	0.846	0.000
S1	on	2.97	0.58	8.74	8.74	0.807	0.807	0.000
S1	off	2.97	0.58	7.95	7.70	0.832	0.833	0.000
S2	on	6.32	1.71	5.69	5.59	0.589	0.590	+0.001
S2	off	6.32	1.71	5.23	5.23	0.599	0.599	0.000

Note: Budget is $L_{cap} - L_0$ in sones; NAL-NL2 use is the loudness NAL-NL2 adds above L_0 . Sones are the aided loudness reached at each floor; SII is the complete desensitized SII. No band received negative gain in any solution.

Search Variability

The optimizer's random-number seed is fixed by the audiogram and level, so repeated runs are identical. To estimate how much a solution depends on the random starts, the floor comparison (anchor on) was repeated with the default seed and five further seeds for the four 1000 Hz edge audiograms at the 0.5-sones budget and for N1, N2, S1, and S2 at NAL-NL2 loudness (Table S4). The default-seed runs reproduced the stored solutions exactly. The pooled within-case

standard deviation of the floor effect was 0.0037 (2 SD = 0.0075). Of the 48 pairs, 40 matched in loudness to within 0.01 sones. In five of the remaining eight the –10 dB solution was the quieter (by up to 0.17 sones), so their floor effects, if anything, understate the benefit; in the other three it was louder by at most 0.016 sones.

Table S4. Complete-SII Floor Effect across Six Random Seeds (Anchor Penalty On)

Case	Seed 0	1	2	3	4	5	Mean	SD
1000 Hz, 20 dB/oct	+0.011	+0.014	+0.005	+0.015	+0.010	+0.011	+0.011	0.004
1000 Hz, 30 dB/oct	+0.015	+0.013	+0.012	+0.018	+0.020	+0.016	+0.016	0.003
1000 Hz, 40 dB/oct	+0.014	+0.016	+0.015	+0.016	+0.013	+0.013	+0.015	0.001
1000 Hz, 50 dB/oct	+0.015	+0.014	+0.015	+0.014	+0.019	+0.014	+0.015	0.002
N1	–0.001	–0.001	–0.007	+0.009	+0.010	+0.003	+0.002	0.007
N2	–0.002	–0.010	–0.005	+0.002	–0.002	+0.007	–0.002	0.006
S1	–0.004	–0.005	–0.002	–0.003	–0.001	–0.002	–0.003	0.002
S2	+0.001	–0.002	0.000	–0.001	0.000	–0.002	–0.001	0.001

Note: The 1000 Hz cases are sweep audiograms at the 0.5-sone budget; N1 to S2 are at NAL-NL2 loudness. Seed 0 is the package default; all runs used 20 starts, so the seed-0 value for the 50 dB/octave case differs slightly from the 40-start value in the main text.

S5. Reproducibility

All analyses were run in R from the repository cited in the Data Availability section. Table S5 lists the script that produces each reported result; all scripts are in `reproducibility_scripts/` and are run from the repository root.

Table S5. Scripts Producing Each Result

Result	Script (in order of execution)
Table 1 audiograms; NAL-NL2 gains	bisgaard_profiles.R (data; gains exported from NAL-NL2 v2.0, dll v2.15)
Table 2; Figure 1; NAL-NL2 loudness	rerun_bisgaard_profiles.R; fig_jaaa.R
Tables 4 and 5; Table S2	rerun_bisgaard_profiles.R
Table 3; resolution limit	rerun_complete_objective.R
Figure 2	fig_jaaa.R (after rerun_complete_objective.R)
Low- vs high-frequency rescoring; unspent budget	mechanism_check_complete.R
Table S1	rerun_ansi_objective.R
Tables S3 and S4; search variability; floor at the normative ceiling	floor_robustness.R

References

Majdak P, Hollomey C, Baumgartner R. (2022) AMT 1.x: A toolbox for reproducible research in auditory modeling. *Acta Acustica*. 6:19. doi:10.1051/aacus/2022011.